JACOB SHKROB

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Education

New York University **2022 – 2027 (expected)**
Ph.D. Mathematics

McGill University **2018 – 2022**
B.Sc. Honors Probability & Statistics, Minor in Computer Science *CGPA: 3.92/4.00*

Honors & Awards

Charles Newman Fellowship **2025**
New York University — awarded to an outstanding graduate student in probability

Finalist, Azrieli Visiting Ph.D. Fellowship **2026**
Azrieli Foundation

1st Place & $25,000, ProjectX Undergraduate AI Research Competition **2022**
Clinical practice cohort — personalizing mechanical ventilation via deep conservative reinforcement learning

Faculty of Science Merit Scholarship **2019 – 2021**
McGill University — awarded in both 2019–2020 and 2020–2021

Dean's Honors List **2019 – 2021**
McGill University — awarded in both 2019–2020 and 2020–2021

Winner, PharmaHacks (Novartis clinical trial data track) **2019**
Montreal Institute for Learning Algorithms (MILA)

Publications & Preprints

[1] J. Niles-Weed, S. Sadovsky, and **J. Shkrob**. An optimal constant for vector balancing with permutations. *arXiv:2610.02127*, 2026.

[2] J. Niles-Weed and **J. Shkrob**. Nearly sharp comparison results for sliced and max-sliced Wasserstein distances. *arXiv:2608.13374*, 2026.

[3] Y. Han, A. Shetty, and **J. Shkrob**. An empirical Bayes perspective on heteroskedastic mean estimation. *arXiv:2603.13499*, 2026.

[4] J. Niles-Weed and **J. Shkrob**. Bayes-optimal estimation for Gaussian planted matchings. In preparation, 2026.

[5] F. Kondrup, T. Jiralerspong, E. Lau, N. de Lara, **J. Shkrob**, M. D. Tran, D. Precup, and S. Basu. Towards safe mechanical ventilation treatment using deep offline reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 15696–15702, 2023.

Research Experience

AstraZeneca *Student Researcher* **May 2022 – Aug 2022**
↪ Dr. Natasha Markuzon, Dr. Damian Bikiel

- Simulated time-to-event data exhibiting both predictive and prognostic behavior. Developed methods for inducing subgroup structure.

McGill Institute for Computational Medicine (MICM) *Summer Research Intern* **May 2020 – Nov 2020**
↪ Prof. Yi Yang *McGill University*

- Developed an algorithm for the time-varying Cox proportional hazards model, with applications to survival analysis and immunology.
- Wrote R packages in C++/R using `Rcpp`, adding parallelism to cut running time.

Big Data Summer Institute (REU) *Student Researcher* **June 2019 – Aug 2019**
↪ Prof. Bhramar Mukherjee *University of Michigan*

- Applied data mining and machine learning to genomic and phenotypic data with the goal of improving drug efficacy for individual cancer patients.

Conferences, Workshops & Summer Schools

Iliad Intensive *Participant (selected, funded)* **October 2026**
↪ Iliad — Berkeley, CA

- Four-week intensive course in technical AI alignment and its mathematical foundations; admission based on mathematical strength.

Princeton Machine Learning Theory Summer School *Participant/Presenter* **August 2026**
↪ Princeton University

- Doctoral training in theoretical machine learning, training dynamics, and diffusion models.

Conference of Learning Theory *Participant* **June 2026**
↪ University of California San Diego

- Presented work on empirical Bayes for heteroskedastic mean estimation.

Workshop on Robust Statistics *Presenter* **April 2026**
↪ Columbia University

- Presented work on empirical Bayes for heteroskedastic mean estimation.

Statistical Optimal Transport Summer School *Participant* **June 2025**
↪ Simons Laufer Mathematical Sciences Institute

- Doctoral training in statistical optimal transport methodology.

Methods in Single-Cell Data Integration and Optimal Transport *Participant* **June 2025**
↪ Irving Institute for Cancer Dynamics

- Multi-modal single-cell integration; emerging spatial and imaging technologies.

Statistical and Numerical Methods for Non-commutative Optimal Transport *Participant* **May 2025**
↪ Institute for Pure & Applied Mathematics (UCLA)

- Numerical algorithms for optimal transport and gradient flows.

Emerging Directions Workshop, NITMB *Participant* **Feb 2025**
↪ University of Chicago

- Discussed open problems in mathematical biology presented by leading researchers.

Probability, Algorithms and Inference *Participant* **Summer 2024**
↪ Georgia Institute of Technology

- Combinatorics of the perceptron model, planted subgraph recovery, sphere packing.

51st Probability Summer School, Saint-Flour *Participant* **Summer 2023**

- Bayesian nonparametric statistics, stochastic homogenization, integrable systems.

Projects & Extracurriculars

Moral Flocking: Do AI Groups Tend Toward Consensus? *Hackathon Project (with R. Budiono)* **Oct 2026**
↪ AI Swarm Dynamics Hackathon — AI Village & Grove Research

- Tested whether LLM agents converge on moral consensus: 12 models from 6 providers debated MoReBench dilemmas over three forum rounds, with private opinions scored by an AI judge.
- Found discussion-driven convergence vs. a control condition, and that collective opinion was robust to over-representation and planted-persuader steering interventions.

PharmaHacks — Novartis Clinical Trial Data *Winning Project* **Fall 2019**
↪ Montreal Institute for Learning Algorithms (MILA)

- Built random forest, naïve Bayes and XGBoost classifiers in `scikit-learn` and `pandas`.
- Used χ^2 tests of independence, ANOVA and bagging/boosting for feature and model selection.

Directed Reading Project *Undergraduate Project* **Fall 2020**

- *Probabilistic analysis of simple random graph models.*

Technical Skills

Programming	Python, R, C++, Java, MATLAB, OCaml, Bash *(learning Julia)*
Tools	LaTeX, PyTorch, `tidyverse`, `ggplot2`, R Markdown, Git/GitHub, Vim, Anaconda, Excel
Methods	Statistical modeling, data wrangling and cleaning, survival analysis, ML pipelines
Languages	English, Russian, French, Hebrew *(basic)*